M5B-E4 / ONE INTEGRATED 144MHz TRUE-AM EVALUATION PCB / NOT RELEASED
PHA-1+ and PD54008L-E retained; documented ST output match and elliptic LPF.
2S9Vboost OR4S9V/7.5Vbuck exclusive populations. Separate3.3Vtraining.
Driver: independent5.456V/5V regulation and18V-rated switch. Power-only qualification beforeR952/MMIC.
Input match has bounded same-board tuning;RF gain/AM/thermal/endurance tests NOT_RUN.
Review-only PCB/Gerbers. No ordering,fabrication,assembly or RF operation authorized.

01 Interface and local SI5351 source

File: 01_interface_source.kicad_sch

02 Minimal independent fail-off and captured ARM context

File: 02_interlocks.kicad_sch

02 Minimal independent fail-off and captured ARM context continued

File: 02_interlocksb1.kicad_sch

03 External protected pack and exclusive conversion populations

File: 03_input_power.kicad_sch

03 External protected pack and exclusive conversion populations continued

File: 03_input_powerb1.kicad_sch

04 2S boost and regulated MMIC rail

File: 04_boost_driver_supply.kicad_sch

05 Sine reconstruction and series AM modulator

File: 05_am.kicad_sch

05 Sine reconstruction and series AM modulator continued

File: 05_amb1.kicad_sch

06 Source filter, economical driver and fixed bias

File: 06_source_driver_bias.kicad_sch

06 Source filter, economical driver and fixed bias continued

File: 06_source_driver_biasb1.kicad_sch

07 Independent very-low-power sine-AM training path

File: 07_training.kicad_sch

07 Independent very-low-power sine-AM training path continued

File: 07_trainingb1.kicad_sch

08 Local voltage, current and thermal trip

File: 08_protection.kicad_sch

08 Local voltage, current and thermal trip continued

File: 08_protectionb1.kicad_sch

06b ST final-stage matching derivative

File: 06b_st_matching.kicad_sch

06c Original ST elliptic harmonic LPF

File: 06c_st_lpf.kicad_sch

10 Local physical switches and clock sequencing

File: 10_enable_detail.kicad_sch

11 External supplies and integrated commissioning

File: 11_external_contract.kicad_sch

12 Independent driver regulator and integrated alignment links

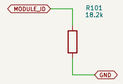
File: 12_m5b_qualification.kicad_sch

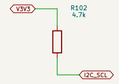
ARDF-TX / NOT RELEASED
Sheet: /
File: RF2M.kicad_sch
Title: ARDF-TX 2 m A2A evaluation schematic
Size: A2 | Date: 2026-10-17 | Rev: M5B-E4
KiCad E.D.A. 10.0.6 | Id: 1/20

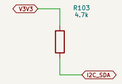
01 Interface and local Si5351 source

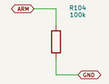
M5B-E4 / Integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

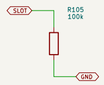


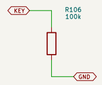


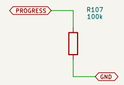


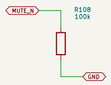


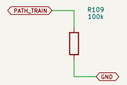


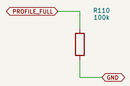


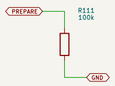


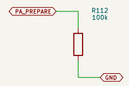


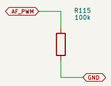


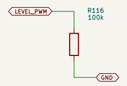




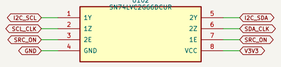


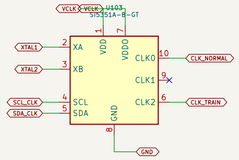


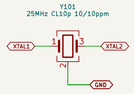


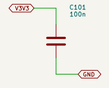


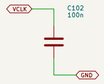


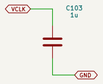
















M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

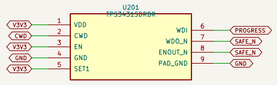
Sheet: /01 Interface and local Si5351 source/
File: 01_Interface_source.kicad_sch

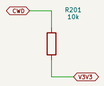
Title: 01 Interface and local Si5351 source

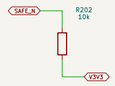
Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 2/20

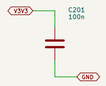
02 Minimal independent fail-off and captured ARM context

M5B-E4 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

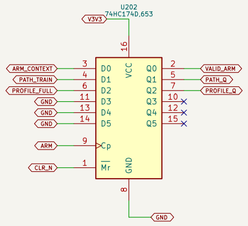














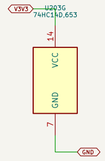






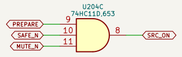


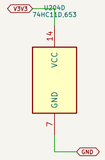


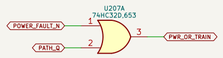
















M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

Sheet: /02 Minimal independent fail-off and captured ARM context/
File: 02_Interlocks.kicad_sch

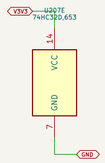
Title: 02 Minimal independent fail-off and captured ARM context

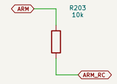
Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 3/20

02 Minimal independent fail-off and captured ARM context continued

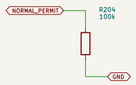
M5B-E4 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

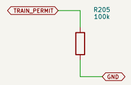




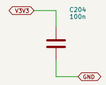


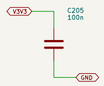


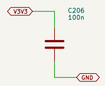


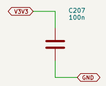


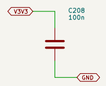


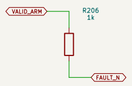


























M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

Sheet: /02 Minimal independent fail-off and captured ARM context continued/
File: 02_Interlocksb1.kicad_sch

Title: 02 Minimal independent fail-off and captured ARM context continued

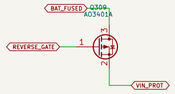
Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 4/20

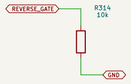
03 External protected pack and exclusive conversion populations

M5B-E4 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN



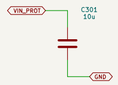


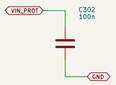


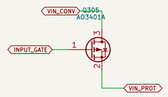


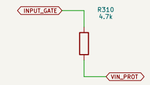


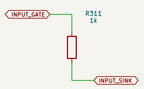


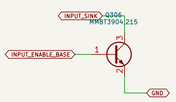


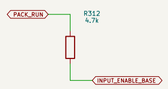


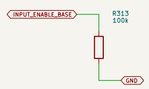






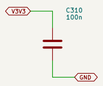




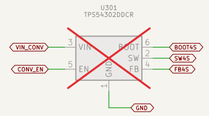










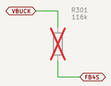




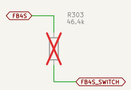


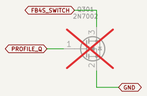


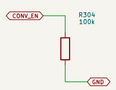


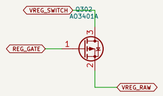


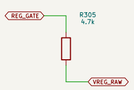












M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

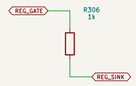
ARDF-TX

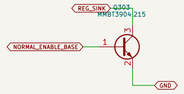
Sheet: /03 External protected pack and exclusive conversion populations/
File: 03_Input_power.kicad_sch

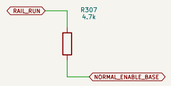
Title: 03 External protected pack and exclusive conversion populations

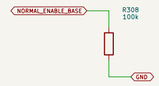
Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 5/20

M5B-E4 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

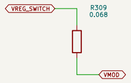






















M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

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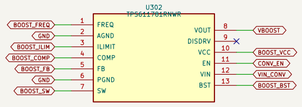
Sheet: /03 External protected pack and exclusive conversion populations continued/
File: 03_input_powerb1.kicad_sch

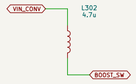
Title: 03 External protected pack and exclusive conversion populations continued

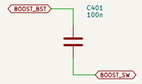
Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 6/20

04 2S boost and regulated MMIC rail

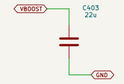
M5B-E4 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN



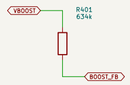


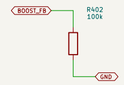


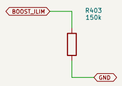


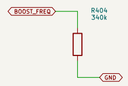


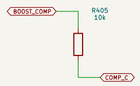


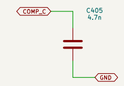


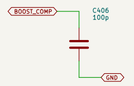


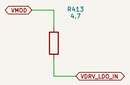


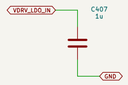




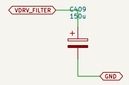
















M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

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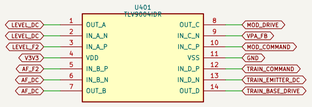
Sheet: /04 2S boost and regulated MMIC rail/
File: 04_boost_driver_supply.kicad_sch

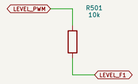
Title: 04 2S boost and regulated MMIC rail

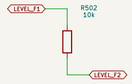
Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 7/20

05 Sine reconstruction and series AM modulator

M5B-E4 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

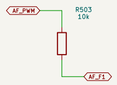


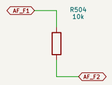






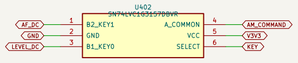


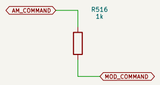


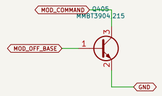


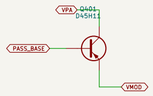


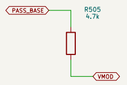


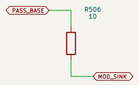


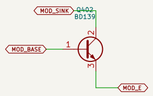


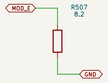


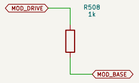


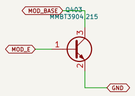


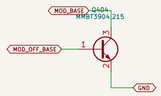


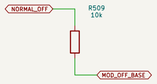


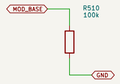


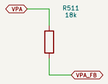


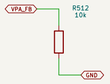


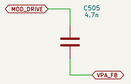




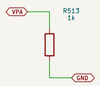




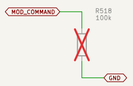


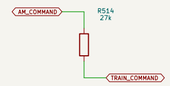












M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

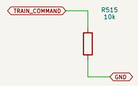
Sheet: /05 Sine reconstruction and series AM modulator/
File: 05_am.kicad_sch

Title: 05 Sine reconstruction and series AM modulator

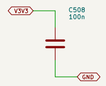
Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 8/20

05 Sine reconstruction and series AM modulator continued

M5B-E4 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN













M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

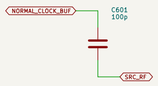
Sheet: /05 Sine reconstruction and series AM modulator continued/
File: 05_amb1.kicad_sch

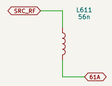
Title: 05 Sine reconstruction and series AM modulator continued

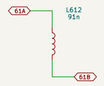
Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 9/20

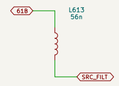
06 Source filter, economical driver and fixed bias

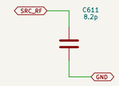
M5B-E4 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN





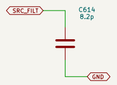


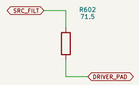


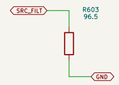


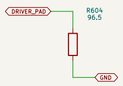


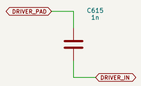


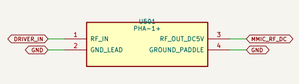


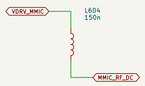


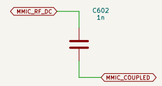


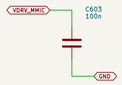


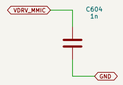


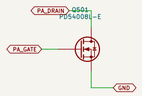


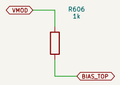


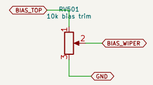




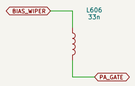


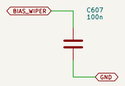


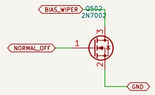














M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

Sheet: /06 Source filter, economical driver and fixed bias/
File: 06_source_driver_bias.kicad_sch

Title: 06 Source filter, economical driver and fixed bias

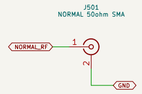
Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 10/20

06 Source filter, economical driver and fixed bias continued

M5B-E4 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN









M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

Sheet: /06 Source filter, economical driver and fixed bias continued/
File: 06_source_driver_biasb1.kicad_sch

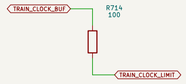
Title: 06 Source filter, economical driver and fixed bias continued

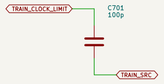
Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 11/20

07 Independent very-low-power sine-AM training path

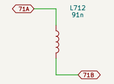
M5B-E4 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

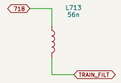


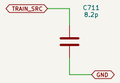


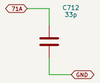


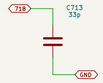


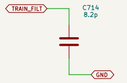


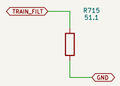


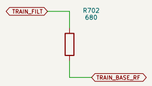


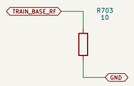


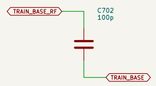


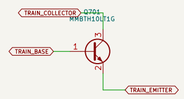


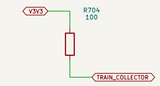


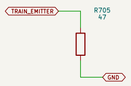


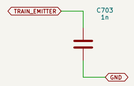


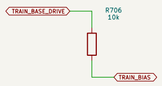


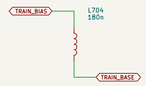


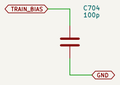


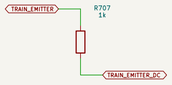


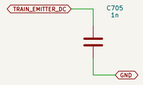


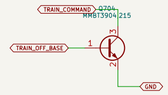


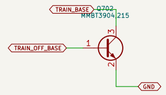


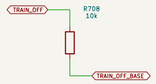


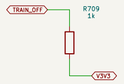


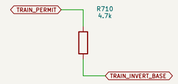


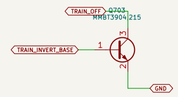


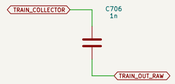












M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

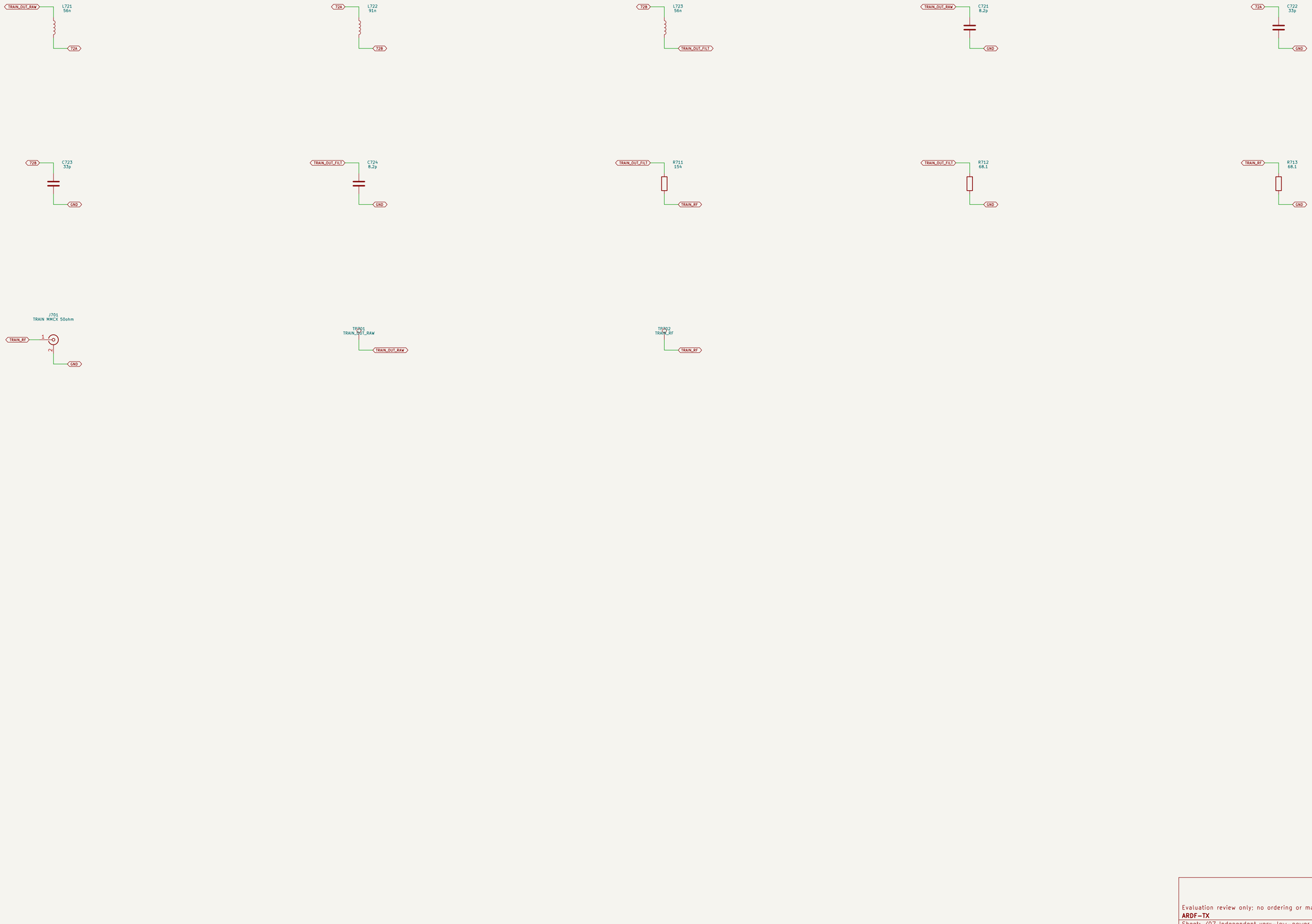
Sheet: /07 Independent very-low-power sine-AM training path/
File: 07_training.kicad_sch

Title: 07 Independent very-low-power sine-AM training path

Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 12/20

07 Independent very-low-power sine-AM training path continued

M5B-E4 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN



M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

Sheet: /07 Independent very-low-power sine-AM training path continued/
File: 07_trainingb1.kicad_sch

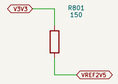
Title: 07 Independent very-low-power sine-AM training path continued

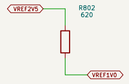
Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 13/20

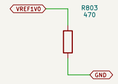
08 Local voltage, current and thermal trip

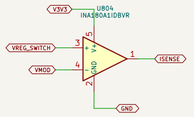
M5B-E4 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

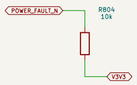


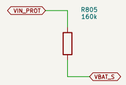


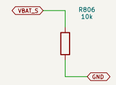


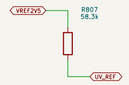


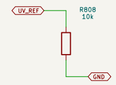


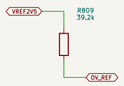


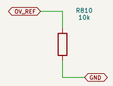


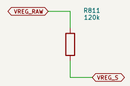


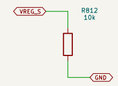


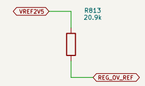


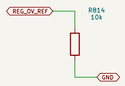


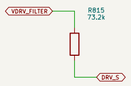


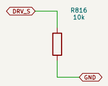


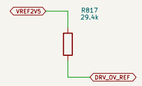


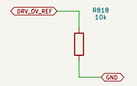


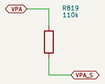


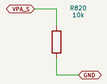


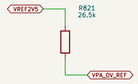


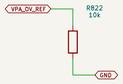


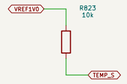


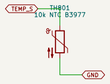


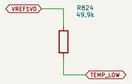


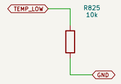


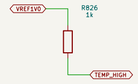


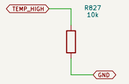












M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

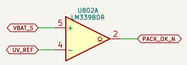
ARDF-TX

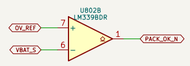
Sheet: /08 Local voltage, current and thermal trip/
File: 08_protection.kicad_sch

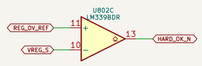
Title: 08 Local voltage, current and thermal trip

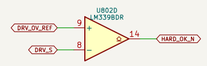
Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 14/20

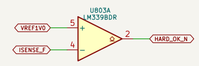
M5B-E4 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

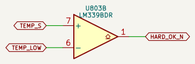


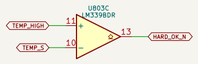


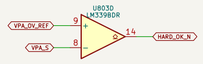


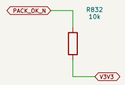


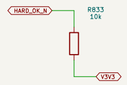








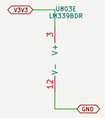


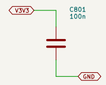


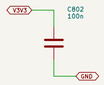


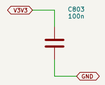


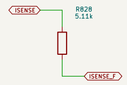


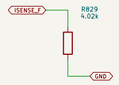


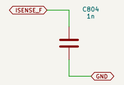






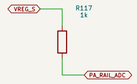


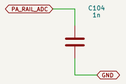














M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

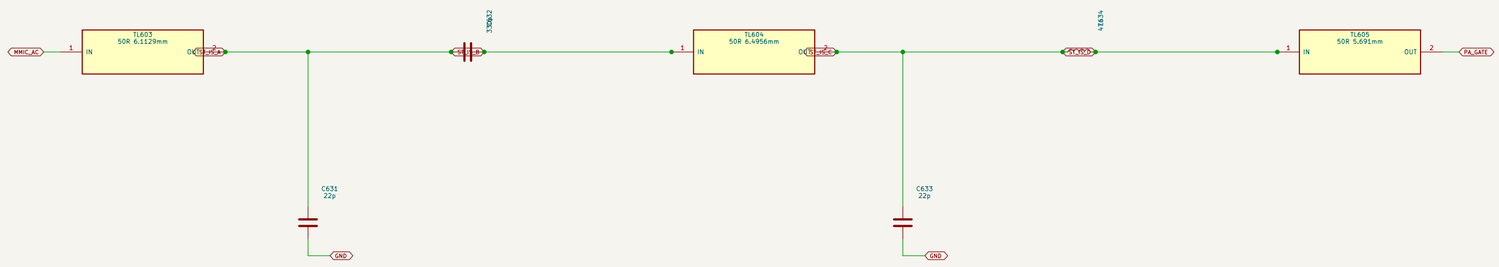
Sheet: /08 Local voltage, current and thermal trip continued/
File: 08_protectionb1.kicad_sch

Title: 08 Local voltage, current and thermal trip continued

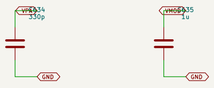
Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 15/20

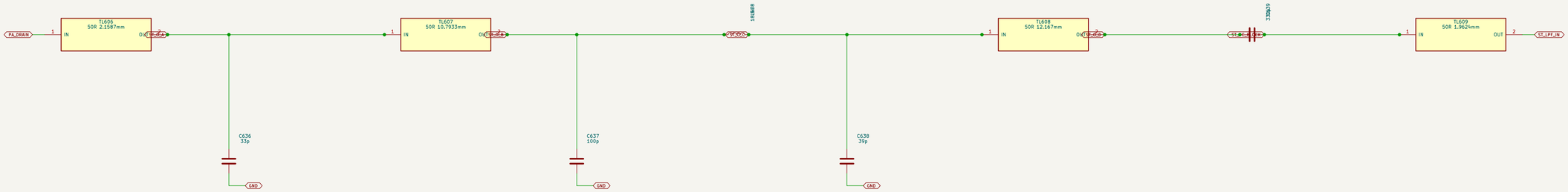
06b ST final-stage matching derivative

ST DocID17343 Rev1 Figs16/17/14. Traced topology; changed MMIC drive and supplyAM remain UNQUALIFIED.
TLine symbols are PCB microstrip lengths, NOT purchased components or zero-length wire. No hardware measurements exist.









Input matching: original interstage source was PD84001 drain, not a50ohm MMIC. No claim of unchanged match or sufficient gain.
Output matching/LPF retained. L633J3s drain feed is a disclosed substitution for the contradictory35.5n/22n ST BOM entry.
Use one integrated end-to-end commissioning procedure after independent electrical/layout review. No separate PA coupons.

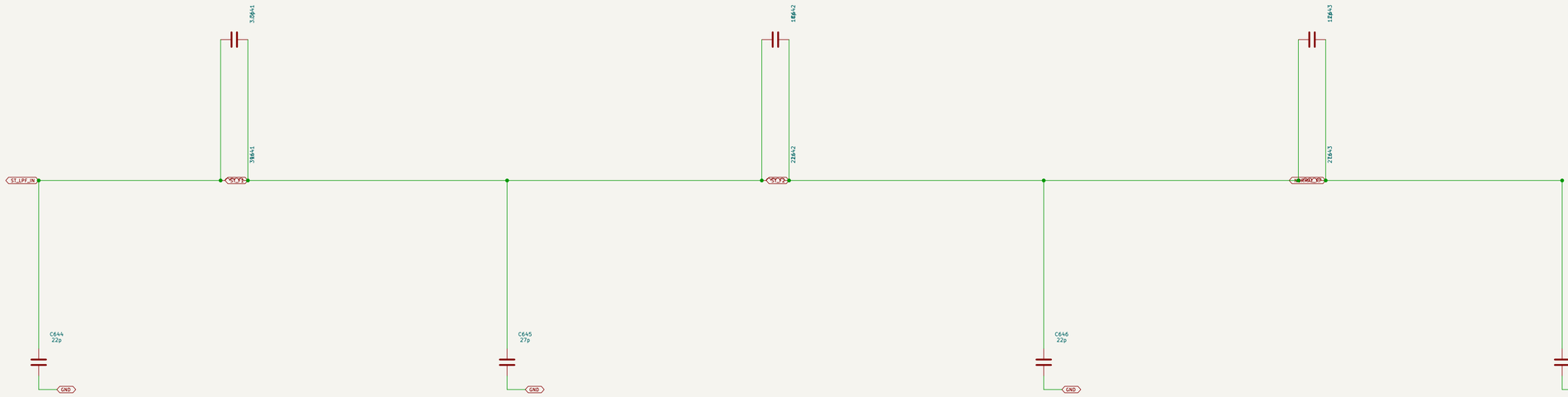
M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX	
Sheet: /06b ST final-stage matching derivative/	
File: 06b_st_matching.kicad_sch	
Title: 06b ST final-stage matching derivative	
Size: A2	Date: 2026-10-17
KiCad E.D.A. 10.0.6	**Rev: M5B-E4**
	Id: 16/20

06c Original ST elliptic harmonic LPF

ST DocID17343 Rev1 Figs16/17/14. Traced topology; changed MMIC drive and supplyAM remain UNQUALIFIED.
TLine symbols are PCB microstrip lengths, NOT purchased components or zero-length wire. No hardware measurements exist.



Input matching: original interstage source was PD84001 drain, not a50ohm MMIC. No claim of unchanged match or sufficient gain.
Output matching/LPF retained. L63533s drain feed is a disclosed substitution for the contradictory35.5n/22n ST BOM entry.
Use one integrated end-to-end commissioning procedure after independent electrical/layout review. No separate PA coupons.

M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

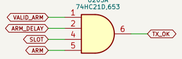
Sheet: /06c Original ST elliptic harmonic LPF/
File: 06c_st_lpf.kicad_sch

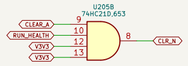
Title: 06c Original ST elliptic harmonic LPF

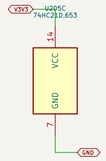
Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 17/20

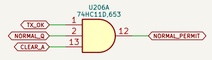
10 Local physical switches and clock sequencing

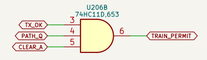
M5B-E4 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN



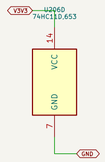




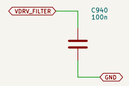


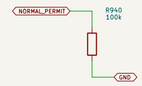


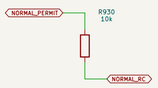


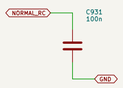


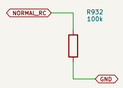




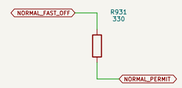


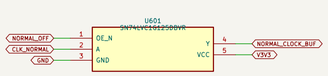


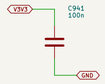














M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

Sheet: /10 Local physical switches and clock sequencing/
File: 10_enable_detail.kicad_sch

Title: 10 Local physical switches and clock sequencing

Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 18/20

11 External supplies and integrated commissioning

M5B-E4 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN





















M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

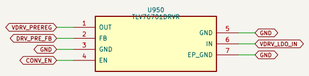
Sheet: /11 External supplies and integrated commissioning/
File: 11_external_contract.kicad_sch

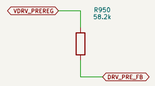
Title: 11 External supplies and integrated commissioning

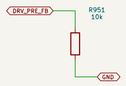
Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 19/20

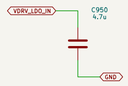
12 Independent driver regulator and integrated alignment links

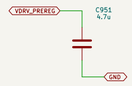
Pre-regulation5.456V then existing5V LDO. Rated18V switch. No20us RC assumption is claimed as protection proof.
Fit driver link only after power-only qualification. RF link and optional50ohm load permit alignment on THIS board.

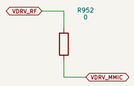


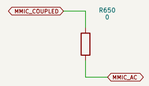


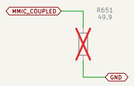






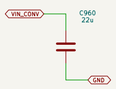


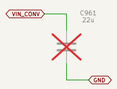












M5B-E4: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

Sheet: /12 Independent driver regulator and integrated alignment links/
File: 12_m5b_qualification.kicad_sch

Title: 12 Independent driver regulator and integrated alignment links

Size: A2	Date: 2026-10-17	**Rev: M5B-E4**
KiCad E.D.A. 10.0.6		Id: 20/20